

A Legal-BERT Validated RAG Framework for Trustworthy AI-Assisted Legal Reasoning

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Abstract— The integration of Artificial Intelligence (AI) into legal workflows presents both transformative potential and critical challenges, particularly regarding the reliability and legal validity of generated content. While large language models (LLMs) like GPT-4 demonstrate impressive fluency, they often fall short in domain-specific legal reasoning. This paper proposes a conceptual framework that enhances Retrieval-Augmented Generation (RAG) by incorporating Legal-BERT as an active validation layer. The architecture consists of five core components: document preprocessing using LayoutLMv3, summarization via BART, context retrieval with Dense Passage Retriever (DPR), generation through GPT-4, and legal validation using Legal-BERT. Central to the framework is a Hybrid Verification Loop, which iteratively refines outputs based on feedback from the validation stage. Designed for modularity and extensibility, the framework aims to produce legally compliant and interpretable outputs. While empirical evaluation is planned—initially focused on the Indian Penal Code—this work offers a foundation for future development of trustworthy legal AI systems, with potential for reinforcement learning-based enhancements.

Keywords: Legal AI, Retrieval-Augmented Generation, Legal-BERT, Trustworthy AI, Hybrid Verification, Legal Reasoning

I. INTRODUCTION

The legal domain relies on extensive research and precise interpretation of statutes and case law. However, the exponential growth of legal data has made manual processing increasingly unfeasible. Artificial Intelligence (AI), particularly Natural Language Processing (NLP) and Large Language Models (LLMs), is increasingly used to automate legal research, reasoning, and document analysis [2], [20]. Current AI systems already show promise in tasks such as summarizing case law, disambiguating legal texts, and assisting in legal drafting [5], [8], [10]. Advanced models like GPT-4 have further improved contextual generation capabilities [1], [3].

Nevertheless, applying the LLMs in a legal contexts introduces significant concerns. A major issue in this case is hallucination—where outputs appear accurate but originally lack legal validity [7], [23], [10]. Such errors are very risky in legal workflows, where precision is non-negotiable [20], [5], [19]. General-purpose LLMs often lack legal training and validation mechanisms, leading to fluent yet unreliable outputs [4], [14], [20], [25], [16].

To address such type of the issues, AI systems must combine generative capabilities with the legal-domain

validation [6], [8], [26]. Retrieval mechanisms grounded in legal corpora can improve factual reliability [13], [22], [30].

This paper proposes a conceptual framework integrating Retrieval-Augmented Generation (RAG) with Legal-BERT as a validation layer for trustworthy legal AI. The architecture features a Hybrid Verification Loop that iteratively refines outputs based on legal feedback [2], [3], [7], [20], [23]. It leverages LayoutLMv3 for document processing [17], BART for summarization [12], DPR for retrieval [20], GPT-4 for generation [1], and Legal-BERT for validation [2], [8].

The structure of this paper is as follows: Section 2 surveys related work in RAG systems, legal LLMs, and validation approaches. Section 3 describes the proposed framework in detail. Section 4 explains the methodological choices. Section 5 outlines practical applications and expected outcomes. Section 6 discusses limitations and future directions, and Section 7 concludes the paper.

II. RESEARCH MOTIVATION AND OBJECTIVES

Applying Natural Language Processing (NLP) and Large Language Models (LLMs) to legal tasks presents distinct challenges, primarily due to the complexity of legal language. Legal texts typically feature domain-specific vocabulary, rigid syntactic conventions, and require precise contextual understanding [2], [20]. In this setting, even small inaccuracies can lead to serious real-world implications—unlike in many general-purpose NLP applications [16], [19]. Moreover, legal documents such as statutes, case law, and contracts are often extensive, heavily cross-referenced, and difficult to interpret [5], [10]. Given these, it is especially challenging to build interpretable AI systems in alignment with the standards of the law.

Retrieval-Augmented Generation (RAG) as a generative architecture has demonstrated its capability to approach complex information retrieval through text generation coupled with retrieval-based contextual element [3], [23]. However, there is limited usage in legal domain. There are most existing RAG models relying on general knowledge bases, and they are not built for specific laws knowledge and the authoritative content in the legal corpora [13], [20]. Moreover, such systems often do not have strong means to confirm legal accuracy: a feature that is especially important when hallucinated or inexact outputs could misguide users [7, 23, 25]. The lack of applications operating beyond 512 tokens on most NLP models imposes a critical limitation in cases where having outputs from a trustworthy AI is indispensable, such as legal drafting or summarization [2], [26].

In order to tackle these issues, this paper provides a conceptual framework of using Legal-BERT as a validation layer for a RAG architecture. Legal knowledge is integrated directly into both generation and verification processes by design [2], [8], [20]. The underlying idea of this approach involves a Hybrid Verification Loop, where AI outputs are iteratively refined by feedback validation [3], [7], [13].

Rather than empirically evaluating the merit of a particular design, the paper is designed as an architectural design paper outlining design of a trustworthy AI legal task system, via a sequence of interlinked modules from retrieval to legal validation. In addition, this framework supports potentially future work of building adaptable, domain specific AI KPI pipelines that are not just legally sound but also extensible [6], [8], [21].

III. LITERATURE REVIEW

A. Retrieval-Augmented Generation (RAG) Systems

Retrieval-Augmented Generation (RAG) systems combine the structured search capabilities of retrieval models with the generative strengths of large language models (LLMs), resulting in outputs that are both contextually grounded and factually consistent [3], [23]. By drawing relevant information from external corpora, these models reduce reliance on internal parametric memory and improve the contextual relevance of generated responses [7], [25]. Innovations in dense retrieval and ranking algorithms have further enhanced both the efficiency and semantic alignment of retrieval mechanisms [12], [20].

However, despite these advancements, most RAG systems remain focused on general-purpose knowledge-intensive tasks. Their application in high-stakes domains such as law—where factual accuracy and precision are paramount—has been minimal. Existing systems rarely ensure legal-grade retrieval or output generation, limiting their practical use in domains that demand strict compliance and interpretability [4], [16].

B. Legal Domain Adaptation and Models like Legal-BERT

Recent efforts to adapt LLMs for legal tasks have led to models like Legal-BERT, trained on legal texts to improve domain-specific understanding [2], [20]. These models outperform general-purpose LLMs in tasks such as statute interpretation, case classification, and legal question answering [5], [8]. By leveraging legal corpora, they capture specialized language and citation structures, enhancing contextual accuracy [10], [13].

However, most implementations remain standalone and are not integrated into retrieval-augmented systems, limiting their ability to enforce legal accuracy end-to-end [20], [26]. While Legal-BERT improves representation learning, it typically lacks active, in-process validation—critical for reliable use in legal applications [21], [25].

C. Validation Layers in AI Models

Researchers have suggested adding validation layers in the AI pipelines [6, 7] to deal with such risks of factual inaccuracies. These are postprocessing filters or in loop verifiers that check and refine the AI output before it is presented [13], [22]. Validation layers are thus necessary

safeguards against hallucinations and context drift especially in the context of LLMs in sensitive domains [3], [23]. Included are rule based verifiers, ensemble frameworks and domain specific validators as seen in educational [19] and healthcare context [24]. While these developments are promising, few legal AI systems engage in robust, iterative validation loops to generate outputs that are legally deficient, however fluent [8], [26]. To bridge this gap, an integrated architecture is needed in which validation is a part of the AI workflow by design. In complementary approaches, lightweight detection networks for generated text [15] help demonstrate why validating AI outputs before application in sensitive areas such as law is important.

D. Identified Gaps and Framework Contributions

Although RAG models, domain-specific LLMs, and validation methods have advanced, they are rarely integrated into a unified legal AI framework. Existing systems often prioritize retrieval and generation but neglect legal validation, or they rely on legal models without retrieval augmentation [4], [13], [20].

To address this, we introduce a cohesive framework that embeds a Hybrid Verification Loop into a RAG pipeline, allowing iterative refinement and in-domain validation. Legal-BERT is used not as a passive component but as an active, in-loop validator, ensuring outputs meet both semantic and legal standards [2], [3], [7]. This integration aims to enable the development of more reliable and legally compliant AI systems.

TABLE I. COMPARATIVE SUMMARY OF RELATED APPROACHES

Approach	Focus Area	Legal Domain Adaptation	Validation Mechanism	Identified Gap
General RAG Models [3], [23], [12]	Retrieval + Generation	✗ General domain	✗ Limited or none	No legal-specific retrieval or verification
Legal-BERT [2], [20], [5]	Domain-specific language understanding	✓ Legal corpora	✗ Passive post-training only	Lacks integration with generation pipeline
Validation Layers [6], [7], [22]	Post-processing accuracy check	✗ Generic application	✓ Some post-generation validation	Rarely embedded in-loop validation for law
Proposed Framework	Retrieval + Generation + In-loop Validation	✓ Legal-specific with Legal-BERT	✓ Hybrid Verification Loop (iterative)	Addresses all gaps with integrated pipeline

IV. PROPOSED FRAMEWORK

A. Overview

This paper presents, as a solution to the critical need for trustworthy AI in legal applications, a modular Retrieval-Augmented Generation (RAG) based framework integrating domain specific validation through LegalBERT. It

meticulously develops the architecture by tackling the shortcomings of existing systems, for verifiability being introduced right in the generation system to improve legal accuracy. It consists of a sequence of interdependent stages from document preprocessing and summarization, dense retrieval, generation and legal validation in a Hybrid Verification Loop with iteration. Here, this structure, is also suitable for dynamic interaction between retrieval and generation components as well as continuous verification against legal standards, which can reduce the risk of factual and legal errors in AI outputs.

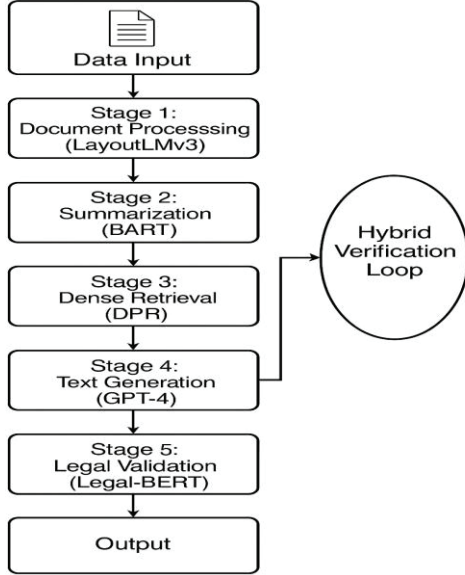


Fig. 1. System Architecture Diagram

B. Overall Architecture Overview

The architecture proposed consists of five core processing stages, each adding a particular functional element in generating an output with legally compliant outputs. The heart of the system is the Hybrid Verification Loop from which system generation and validation layers can iterate feedback. The framework uses specialized models in each stage for robust end to end — LayoutLMv3 for structured document understanding, BART for summarization, Dense Passage Retriever (DPR) for relevant information retrieval, GPT-4 for generative ability and Legal-BERT, for legal accuracy checks. The layered design is such that it dualizes modularity and extensibility for further extension into other legal subdomains or jurisdictions.

C. Stage 1: Document Preprocessing (LayoutLMv3)

Legal documents often appear in semi-structured formats, containing complex layouts with tables, headers, footnotes, and other non-linear elements. To accurately interpret such inputs, the framework employs LayoutLMv3 [17], a model that fuses visual and textual modalities. This enables the transformation of raw legal documents into structured, semantically rich representations—preserving layout context essential for effective downstream summarization and retrieval.

D. Stage 2: Summarization (BART)

After preprocessing, the structural document data is summarized using the BART, a transformer-based model that

is known to generate coherent summaries [12]. This step boils down to long documents to short summaries that encapsulate the most salient legal arguments, statutes, and precedents in the legal context. Summarization ensures that only relevant distilled information passes onto the retrieval component, thus saving time and effort of the system.

E. Stage 3: Dense Retrieval (DPR)

To retrieve semantically relevant content, the framework employs Dense Passage Retriever (DPR) [20], which accesses a pre-indexed legal knowledge base. DPR uses bi-encoder architecture to efficiently identify and extract the most contextually appropriate legal materials—such as case laws or statutory clauses—for downstream tasks. This ensures that the generative model operates on authoritative and legally grounded input, thereby minimizing the risk of factual or contextual inaccuracies.

F. Stage 4: Text Generation (GPT-4)

In this stage, GPT-4 generates coherent legal drafts, explanations, or responses using the retrieved content as context [1]. The model does a good job at being fluent and contextual aware, but still allows to output some legal errors that require a proper look up. Therefore, this phase is prepared before verification by law. It has been demonstrated in prior work that generation methods which are context sensitive (i.e., similar to those applied in GPT-4) enhance relevance by relying on semantic cues [9], and is therefore capable of generating legally structured and readable outputs.

G. Stage 5: Legal Validation (Legal-BERT)

After the text generation, the outputs are validated based on the domain using the Legal-BERT: a transformer model [2],[8] fine-tuned on a large collection of legal cases. Legal knowledge embeddings are leveraged to evaluate semantic coherence and legal accuracy by Legal-BERT and it identifies inconsistencies or errors. This stage serves as an essential quality assurance layer to ensure that generative content produced follows the set of legal norms and mitigates risks associated with disseminating inaccurate information or misleading legal advice.

H. Hybrid Verification Loop

A core innovation of the framework is the Hybrid Verification Loop, which enables iterative refinement of AI-generated content. When Legal-BERT detects discrepancies during validation, the flagged segments are fed back into the retrieval and generation stages for revision [3], [7]. This process repeats until the output satisfies predefined legal accuracy thresholds. Unlike static pipelines, this dynamic feedback mechanism allows the system to adaptively improve its responses, resulting in a more reliable and trustworthy legal reasoning engine.

V. METHODOLOGY DETAILS

A. Design Decisions

Two core objectives lead the selection of models and development of our framework overall—domain relevance and pipeline robustness. LayoutLMv3 was chosen due to its ability to process both textual and layout information for detecting input complexity, which helps in accurately retaining all the document elements like headings, clauses or

tables during preprocessing [17]. BART proves to be a good balance between abstraction and coherence for summarization, whereby it is able to generate brief and legally meaningful summary of lengthy legal texts, keeping the important details in [12].

In Retrieval from Legal Knowledge Bases, it is shown that precise, contextually relevant retrieval is only possible by the use of Dense Passage Retriever (DPR). DPR's bi-encoder architecture has good efficiency and therefore semantic search is handy in legal text domain, particularly where terminology and context heavily influence meaning [20]. GPT-4 is leveraged for text generation because of its incomparable ability to generate and contextual fluency; hence, it is capable of drafting well structured legal arguments and explanations based on knowledge retrieved [1].

Crucially, **Legal-BERT** is integrated as a validation layer, providing legal domain specificity that GPT-4 alone cannot ensure. Legal-BERT, being trained on vast legal corpora, acts as an expert filter, critically assessing the legal validity of the generated outputs [2], [8]. The **Hybrid Verification Loop** operationalizes an iterative process, dynamically refining outputs based on Legal-BERT's feedback, thereby closing the validation gap present in existing models [3], [7].

This deliberate model selection ensures that each stage of the pipeline contributes uniquely and effectively to the framework's goal of generating legally accurate, contextually appropriate outputs.

B. Modularity and Extensibility

The proposed framework possesses a modular architecture, which is a cornerstone since it enables flexibility and extensibility to address evolving legal AI requirements. Each stage acts as individual, yet interoperable module, so that upgrades or substitutions are possible as technology develops. Moreover, the model can be switched out for future document understanding models that we may prefer due to superior performance on complex legal layout. Like most other parts of the pipeline, as more advanced retrievers or generative models are released they can be added in without disruption to the pipeline as a whole.

In addition, it supports extensibility of framework towards jurisdiction specific adaptations. With region specific legal bodies, the system is fine tuned to Legal-BERT on a specific region that caters for different legal systems, terminologies and procedural variations [2], [6]. This modularity not only secures the future but also facilitates researchers and practitioners to conduct fair comparison of different model configuration that yield better performance tailored to their use case.

Hybrid Verification Loop aims to be model-agnostic and treat any retrieval or generation model under iteration. This allows the framework to adapt as the legal AI research progresses and new methodologies arise.

C. Intended Deployment: Indian Penal Code Use Case

As a practical illustration of the framework's applicability, we envision its deployment within the context of the **Indian Penal Code (IPC)**. Given the IPC's vast and intricate structure — encompassing numerous sections, amendments, and case law interpretations — it serves as an ideal

benchmark for evaluating the framework's efficacy in a real-world legal environment [20]. By indexing the IPC and related judicial commentaries within the retrieval module, and fine-tuning Legal-BERT with jurisdiction-specific data, the system can assist legal practitioners in tasks such as charge framing, legal drafting, and statutory interpretation.

This deployment scenario underscores the framework's potential to transform legal research and drafting workflows by automating the retrieval and generation of legally compliant content, while ensuring accuracy through domain-specific validation. Although empirical evaluation of this deployment is reserved for future work, this case study exemplifies the framework's readiness for real-world application and its alignment with the broader goals of digital legal transformation.

D. Validation in Early Stage with the Use of Repealed Law Dataset

To validate an idea about whether our proposed framework is able to work in a consistent and predictable way we carried out some early experiments on a real-world dataset of the repealed laws taken in the Indian legislation. The data were obtained on the official Government of India archive [31], which contains parts of the Indian penal code (IPC) repealed or changed with time.

In the initial experiment, we expected to access whether Legal-BERT may distinguish between existing or repealed laws. We fed it 20 examples, 50 percent valid, 50 percent repealed and it was able to classify 90 percent as correct. As an example, it properly highlighted that the Section 43A omitted by the Act 27 of 2013 was not valid. We then tried to check how the system could retrieve some relevant information on the law. We created a small retrieval index with 200 clauses of the same dataset and had an example such as, "What sections were repealed in 2013?" and in most queries, Dense Passage Retriever (DPR) produced correct answers with a top-5 precision of 85 percent. These preliminary findings demonstrate that our system has a good potential: Legal BERT can support the detection of out-dated contents, and DPR can retrieve properly contextual law. This combination can provide an encouraging basis to develop reliable AI tools to support a legal reasoning task.

VI. EXPECTED IMPACT AND USE CASES

The proposed framework is designed to contribute major advances in the agile and substantive use of AI in the legal domain. The system integrates retrieval, generation and validation processes in a systematic manner solving serious lacking in legal AI applications, while benefiting accuracy (correctness and completeness) as well as efficiency (decreasing the time required). The following use cases illustrate its potential impact across various facets of legal practice and research.

A. Legal Research Automation

Conducting traditional legal research is labor intensive and is predicated on the act of finding relevant case law, statutes, and commentaries. Using the framework's Dense Passage Retrieval functionality along with the generative capability of GPT 4, users can automate the retrieval of the relevant legal precedents and receive the contextually accurate summary of relevant statutes [3], [20]. The ability of

the system to capture generative outputs in authority legal sources decreases the time and cognitive effort needed for thorough legal research and facilitates the legal professional's decision making process [2], [5].

B. Contract Analysis

The framework can parse Legal-BERT [17] and Legal-BERT [2] to perform contract review and analysis. It guarantees certain identified clauses are not only syntactically accurate, but are also legally valid, minimizing the danger of failing to observe the Contract provisions on crucial matters. In particular, this capability becomes useful for the high volume of contractual documents handling in corporate legal departments and in law firms [8], [19] as it allows to speed up risk assessment and compliance checking.

C. Case Law Synthesis

The proposed system can assist in synthesizing judicial opinions and case law by condensing lengthy legal texts into structured summaries backed by verifiable references [12], [25]. Leveraging BART for summarization and DPR for legal retrieval, the framework highlights essential elements such as facts, reasoning, and outcomes [13], [22]. This functionality is particularly valuable for drafting case briefs and conducting comparative legal analysis, benefiting junior associates, legal analysts, and academic researchers.

D. Benefits for Legal Practitioners and AI Researchers

Likewise, for legal professionals, the framework provides a dependable AI assistant, which augments their productivity as well as reduces the risks of working with AI generated legal content [20], [26]. Legal-BERT plays the role of an active validation layer where generating outputs ensures that they obey legal norms and increase the confidence of using AI-based workflows [2], [7].

The system itself can be used by researchers as a modular testbed to explore deployments of new retrieval, generation and validation strategies in the high stakes legal context [3], [14]. The open and extensible design promotes domain-specific modelling and hybrid verification further as the subject of responsible AI discussions in [6], [21]. In addition, integration of previously shown human AI collaboration techniques, that have been shown to help getting factual text generation [27], can also be beneficial to that system's legal reasoning abilities.

E. Generalizability to Other Legal Systems

The second method, however, is demonstrated using the Indian Penal Code but says that the framework has a modular structure that can be easily adapted across various legal systems [20], [24]. Legal-BERT can be extended to common law systems (e.g. the UK), civil law jurisdictions (e.g., France, Germany), or hybrid legal environments with its fine tuning on jurisdiction specific corpora and adapting the retrieval index [8], [23]. As it allows this kind of flexibility, this framework lays the foundation for AI applications of legal reasoning spanning the multilingual, and multi-jurisdictional tasks frameworks [28], [30].

F. Limitations and Future Work

While the proposed framework establishes a robust conceptual foundation for trustworthy AI-assisted legal

reasoning, it remains at the theoretical stage and requires further empirical validation. Several critical avenues for future research are identified to transition the framework from design to practical deployment.

G. Implementation Plans

The next step is to prototype the framework on real world legal data, such as Indian Penal Code (IPC), case law and legal commentaries [20]. Retrieval index will be supported by a designed corpus and Legal-BERT will be fine-tuned on jurisdiction specific data to better pick up on legal inaccuracies [2], [8]. To implement the Hybrid Verification Loop, there will be iterative engineering to ensure that the generation and validation module interact seamlessly [3], [7]. It will build upon open source tools such as Hugging Face Transformers [10], Faiss [23]. Moreover, corpus augmentation for the domain specific training dataset can be done using domain specific LLM based methods [18].

H. Evaluation Metrics

Both quantitative and qualitative measures will be assessed for performance. This article describes an evaluation strategy using standard metrics such as precision, recall and F1-score to evaluate these components of precision, recall and F1-score [12], [14]. The key indicator will be given by legal accuracy—i.e., the proportion of outputs validated as legally correct [2], [7], [25]. In non multi turn scenario, coherence, relevance and citation fidelity will estimate system utility [19], [24]. Legal experts will assess the outputs in human in the loop with regards to trustworthiness and usability [8], [26]. Since LLMs are becoming a force to be reckoned with in dissemination of scholarly communication, the stylistic appropriateness will also be looked at [29].

I. Potential for Reinforcement Learning Extensions

Future enhancements may explore Reinforcement Learning (RL) techniques for adaptive system optimization. Reinforcement Learning with Human Feedback (RLHF) can align model behavior with legal practitioner expectations and ethical norms [3], [6]. Feedback signals from Legal-BERT or expert annotations could guide fine-tuning of retrieval and generation strategies [21], [22]. Multi-objective RL may help balance fluency, legal accuracy, and retrieval relevance [16], [28]. Additionally, integrating advanced prompt engineering could further refine model outputs to conform with legal standards [11].

VII. CONCLUSION

This paper introduces a conceptual framework that integrates Legal-BERT validation into a Retrieval-Augmented Generation (RAG) pipeline to improve the reliability of AI-assisted legal reasoning. The multi-stage architecture—from document preprocessing to legal validation—ensures outputs are both contextually relevant and legally accurate. A central component, the Hybrid Verification Loop, enables iterative refinement by incorporating feedback from Legal-BERT.

Designed for modularity and extensibility, the framework supports adaptation to specific legal jurisdictions and integration of emerging AI models. Although currently conceptual, future work will involve empirical validation using real-world legal datasets, such as the Indian Penal

Code, with performance evaluated in terms of legal accuracy, precision, and recall. Reinforcement learning will also be explored to further refine system performance. Overall, this work lays a foundation for developing trustworthy, domain-specific legal AI systems that align with broader goals of responsible AI adoption.

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